Simon Briend

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Interests

Statistics, probability theory, random trees, random graphs, network archeology, graph inference, data analysis, information propagation.

Education

PhD **Université Paris-Saclay and Universitat Pompeu Fabra**, Mathematics — Sept 2021 – June 2024
Supervised by Gábor Lugosi and Christophe Giraud. *Inference of the past of random structures and other random problems.* pdf

MS **Université Paris-Saclay**, Mathematics for Artificial Intelligence — Sept 2019–July 2021
Master's degree with a focus on statistical learning and high-dimensional statistics

BS **Université Paris-Saclay**, Pure Mathematics — Sept 2018–July 2019
Generalist mathematics training, with a focus on Probability and Statistics

BS **Lycée Thiers, Marseille**, Classes préparatoires MP* — Sept 2016–July 2018
Generalist scientific training with a strong focus on Mathematics and Physics

Professional experience

Unidistance, Postdoctoral researcher in the Mathematics and Computer Science department, under the supervision of David Belius — Sept 2024–Present, Brig, CH

SLMath (MSRI), Postdoc Fellow in the Probability and Statistics of Discrete Structures program — Jan 2025–May 2025, Berkeley, CA

Universitat Pompeu Fabra, Research assistant in the Economics department — Oct 2021–Sept 2024, Barcelona, ES

Universitat Pompeu Fabra, Research internship, supervised by Gábor Lugosi — May 2021–July 2021, Barcelona, ES

Publications and preprints

In all publications, authors are listed in alphabetical order.

Packing and covering random trees Caelan Atamanchuk, *Simon Briend*, Serte Donderwinkel, Elaine Herrera, arXiv: 2610.40150 pdf — 2026

The packing and covering number of trees are local Caelan Atamanchuk, *Simon Briend*, Serte Donderwinkel, Elaine Herrera, arXiv: 2610.37539 pdf — 2026

TAP variational formula for 2-spin models with general interaction matrix and marginality David Belius, *Simon Briend*, Volodymyr Fomichov, arXiv:2610.13670. pdf — 2026

History estimation in random recursive trees: Pointwise approach via iterated Jordan centralities. Johannes Bäumler, *Simon Briend*, Joost Jorritsma, arXiv:2606.24465. pdf — 2026

Does freezing impede the growth of random recursive trees? Anna Brandenberger, *Simon Briend*, Hanna A. Cairns, Robin Khanfir, Igor Kortchemski, arXiv:2505.16571. pdf — 2025

Random Friend Trees. Louigi Addario-Berry, *Simon Briend*, Luc Devroye, Serte Donderwinkel, Céline Kerriou, Gábor Lugosi, Random Structures & Algorithms pdf — 2024

Leaf Stripping on Uniform Attachment Trees. 2025
Louigi Addario-Berry, Anna Brandenberger, *Simon Briend*, Nicolas Broutin, Gábor Lugosi, Random Structures & Algorithms. pdf

Estimating the history of a random recursive tree. 2025
Simon Briend, Christophe Giraud, Gábor Lugosi, Déborah Sulem, Bernoulli. pdf

On the quality of randomized approximations of Tukey's depth. 2025
Simon Briend, Gábor Lugosi, Roberto I. Oliveira, SIAM Journal on Mathematics of Data Science (SIMODS). pdf

Broadcasting in random recursive dags. 2025
Simon Briend, Luc Devroye, Gábor Lugosi, ESAIM Probability and Statistics. pdf

Archaeology of random recursive dags and Cooper-Frieze random networks. 2023
Simon Briend, Francisco Calvillo, Gábor Lugosi, Combinatorics, Probability and Computing. pdf

Talks and invitations

Invited talk at the POPS seminar, Weierstrass Institute, Berlin, DE.	September 2026
Invited talk at the Department of Mathematics and CS seminar, Universität zu Köln, DE.	June 2026
Invited talk at the Department of Economics seminar, Universitat Pompeu Fabra, Barcelona, ES.	April 2026
Invited talk at the Detection, Estimation, and Reconstruction in Networks workshop. SLMath, Berkeley, US.	May 2025
PSDS seminar at SLMath, Berkeley, US.	Feb 2025
Contributed talk at the PSYR seminar, Unidistance, Brig, CH.	Nov 2024
Invited talk at the ASCAI workshop, University of Potsdam, DE	Feb 2024
Invited talk at Bienal RSME, Pamplona, ES.	Jan 2024
Invited talk at the SfS-PhD presentations at ETHZ, Zürich, CH.	Nov 2023
Invited talk at the APS conference, Nancy, FR.	June 2023
Invited participant. Bellairs workshop on Probability and Combinatorics, McGill Bellairs Institute, Holetown, BB.	2023–2026
Two month-long research visits at McGill University, working with Prof Louigi Addario-Berry, Prof Luc Devroye and Prof Serte Donderwinkel. Montréal, CA.	2022 & 2023
Two month-long research visits at IMPA, working with Prof Roberto Imbuzeiro Oliveira. Rio de Janeiro, BR.	2022 & 2023
Invited talk at the LIMDA seminar at Universitat Politecnica de Catalunya, Barcelona, ES.	Nov 2022
Invited talk at the department of Economics seminar, Universitat Pompeu Fabra, Barcelona, ES	Oct 2022
Contributed talk at the RandNET workshop on random graphs, Eindhoven, NL.	Aug 2022

Teaching

Lecturer for the undergraduate course Optimization and Machine Learning, Unidistance, Brig, CH.	2026
Teaching assistant for the undergraduate course Optimization and Machine Learning, Unidistance, Brig, CH.	2025

Mentoring role in the preparation of an undergraduate thesis at Unidistance, Brig, CH. 2025–2026

Mentoring role in the preparation of an undergraduate research seminar at Unidistance, Brig, CH. 2025

Teaching assistant for the undergraduate course Ordinary Differential Equations, Unidistance, Brig, CH. 2025

Teaching assistant for the undergraduate courses Mathematics II and Real Analysis, Universitat Pompeu Fabra, Barcelona, ES. 2024

Funding

Unidistance visibility grant, funding a research visit to Universitat Pompeu Fabra. 2026

Swiss National Science Foundation scientific exchange grant, funding the research visit of Serte Donderwinkel. 2025

Centre de Recherches Mathématiques travel grant, funding a research visit to McGill University. 2023

DimMathInnov PhD grant from Région Île-de-France. 2021–2024

Sophie Germain master's scholarship, from the Fondation Mathématiques Jacques Hadamard. 2019–2021

Academic service

Member of the Math and CS Faculty council at Unidistance. 2026–2027

Member of the Ethics Committee, in charge of overhauling ethics guidelines for researchers at Unidistance. 2025–2026

Founder and organizer of the PSYR seminar at Unidistance, a research seminar geared towards the young researcher community. 2024–2025

Technical skills

- Proficient in Python and PyTorch for training and fine-tuning large language models (LLMs)
- Experienced with high-performance computing (HPC) clusters and remote job scheduling

Languages

- French and English: C2
- Spanish: B1
- German: A2

Citizenship and work status

- French citizen
- Swiss B-permit (work authorized)